

Sara Kangaslahti

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Education

Harvard University

Doctor of Philosophy Candidate in Computer Science

Cambridge, MA

Expected Graduation: May 2028

Harvard University

Master of Science in Computer Science

Cambridge, MA

May 2026

California Institute of Technology

Bachelor of Science in Computer Science

Minor in Information and Data Sciences

GPA: 4.2/4.3

Pasadena, CA

June 2023

Research Experience

Graduate Research Fellow

School of Engineering and Applied Sciences, Harvard University

Advised by Professor David Alvarez-Melis

Cambridge, MA

August 2023 - Present

Undergraduate Researcher

Department of Computing & Mathematical Sciences, California Institute of Technology

Advised by Professor Anima Anandkumar and Professor R. Michael Alvarez

Pasadena, CA

October 2019 - June 2023

Research Interests: science of large language models, model representation geometry, model interpolation, knowledge distillation, training dynamics, interpretability, model pruning, model merging

Publications

Understanding Layer Patching in Model Size Interpolation

Preprint, 2026 [\[link\]](#)

Also presented at the ICML AdaptFM workshop, 2026

Sara Kangaslahti, Jonathan Geuter*, Nihal V. Nayak*, Marco Fumero*, Francesco Locatello, David Alvarez-Melis.

Thinking at the Right Size: Amortized Distillation Across Post-Trained LLMs

Findings of Empirical Methods on Natural Language Processing, 2026 [\[link\]](#)

Yan Zhou, Sara Kangaslahti*, Jonathan Geuter*, Nihal V. Nayak, Marco Fumero, Francesco Locatello, David Alvarez-Melis.

Inverse Depth Scaling From Most Layers Being Similar

International Conference on Machine Learning, 2026 [\[link\]](#)

Yizhou Liu, Sara Kangaslahti, Ziming Liu, Jeff Gore.

Boomerang Distillation Enables Zero-Shot Model Size Interpolation

International Conference on Learning Representations, 2026 [\[link\]](#)

Also presented at the NeurIPS UniReps Workshop, 2025.

Sara Kangaslahti, Nihal V. Nayak*, Jonathan Geuter*, Marco Fumero, Francesco Locatello, David Alvarez-Melis.

Hidden Breakthroughs in Language Model Training

International Conference on Learning Representations, 2026 [\[link\]](#)

Also presented as **Spotlight** at the ICML Workshop on Mechanistic Interpretability, 2024.

Sara Kangaslahti, Elan Rosenfeld, Naomi Saphra.

Activating Identity and Political Action in the #MeToo Era

Computational Communication Research, 2026 [\[link\]](#)

Melina Much, Daniel Ebanks, [Sara Kangaslahti](#), Jean Kossaifi, R. Michael Alvarez, Anqi Liu, Anima Anandkumar.

GenAI for Systems: Recurring Challenges and Design Principles from Software to Silicon

Preprint, 2026 [\[link\]](#)

Arya Tschand, Chenyu Wang, Zishen Wan, Andrew Cheng, Ioana Cristescu, Kevin He, Howard Huang, Alexander Ingare, Akseli Kangaslahti, [Sara Kangaslahti](#), Theo Lebyrk, Hongjin Lin, Jeffrey Jian Ma, Alexandru Meterez, Clara Mohri, Depen Morwani, Sunny Qin, Roy Rinberg, Paula Rodriguez-Diaz, Alyssa Mia Taliotis, Pernille Undrum Fathi, Rosie Zhao, Todd Zhou, Vijay Janapa Reddi.

Continuous Language Model Interpolation yields Dynamic and Controllable Text Generation

Transactions on Machine Learning Research, 2025 [\[link\]](#)

Also presented at the NeurIPS Workshop on Adaptive Foundation Models, 2024.

[Sara Kangaslahti](#), David Alvarez-Melis.

Topic models estimators are statistically inconsistent and substantively misleading: evidence and solutions

Preprint, 2025 [\[link\]](#)

Danny Ebanks, [Sara Kangaslahti](#).

Analyzing Political Text at Scale with Online Tensor LDA

Political Analysis, 2025 [\[link\]](#)

[Sara Kangaslahti](#), Danny Ebanks, Jean Kossaifi, R. Michael Alvarez, Anima Anandkumar.

Fast Forwarding Low-Rank Training

Empirical Methods in Natural Language Processing, 2024 [\[link\]](#)

Adir Rahamim, Naomi Saphra, [Sara Kangaslahti](#), Yonatan Belinkov.

Can you label less by using out-of-domain data? Active & transfer learning with few-shot instructions

NeurIPS Transfer Learning for Natural Language Processing Workshop, 2023 [\[link\]](#)

Rafal Kocielnik*, [Sara Kangaslahti](#)*, Shrimai Prabhumoye, Meena Hari, Michael Alvarez, Anima Anandkumar.

Adaptive Sampling: Algorithmic vs. Human Waypoint Selection

Preprint, 2021 [\[link\]](#)

Stephanie Kemna, [Sara Kangaslahti](#), Oliver Kroemer, Gaurav S. Sukhatme.

Calibrating User Bias In Sentiment Estimation From Social Media

IEEE International Conference on Big Data (Big Data) Workshop on Big Social Media Data Management and Analysis (BSMDMA), 2018 [\[link\]](#)

[Sara Kangaslahti](#), Anoop Kumar.

* Equal contribution.

Fellowships and Awards

NSF Graduate Research Fellow

August 2023 - Present

National Science Foundation

Doris S. Perpall SURF Speaking Competition Finalist

2022

California Institute of Technology

Top 3% undergraduate research presentation

Carolyn Ash Summer Undergraduate Research Fellow (SURF)

June 2022 - September 2022

California Institute of Technology

Jack E. Froehlich Memorial Award Nominee

2022

California Institute of Technology

Top 5% GPA in Caltech junior class

Beckman Leadership Award Nominee

2022, 2023

California Institute of Technology

Summer Undergraduate Research Fellow (SURF)

California Institute of Technology

June 2020 - August 2020

Industry Experience

Research Intern

NVIDIA Research

Mentored by Professor Anima Anandkumar

March 2022 - June 2023

Software Engineer Intern

AI Integrity Team, Facebook

Mentored by Dr. Liang Tan

June 2021 - August 2021

Teaching Experience

Teaching Fellow

School of Engineering and Applied Sciences, Harvard University

Course: Computational Science and Engineering Capstone Project, taught by Dr. Weiwei Pan

September 2024 - December 2024

Teaching Assistant

Department of Computing & Mathematical Sciences, California Institute of Technology

Course: Distributed Systems, taught by Professor Kianianthra M. Chandy

September 2022 - December 2022

Talks

Boomerang Distillation Enables Zero-Shot Model Size Interpolation

Invited Talk at ELLIS x UniReps Seminar Series [[link](#)]

Remote

February 2026

Boomerang Distillation Enables Zero-Shot Model Size Interpolation

Invited Talk at MIT PLS AI Journal Club

Cambridge, MA

November 2025

Hidden Breakthroughs in Language Model Training

Invited Talk at MIT PLS AI Journal Club

Cambridge, MA

October 2025

Analyzing Social Media Conversations at Scale with Tensor LDA

Invited Talk at University of California Los Angeles (UCLA) Biostats Seminar

Los Angeles, CA

April 2022

Reviewing

Reviewer for ICLR (2025, 2026), NeurIPS (2026), JMLR (2022)

Outreach

- **Volunteer Math and Science Tutor**, Caltech Rise Program, September 2022 - June 2023
- **Volunteer Math and Science Tutor**, Youth of Promise Program, June 2022 - July 2023
- **Volunteer Math Tutor**, NCNW Young Legends Program, January 2022 - May 2022

Language Skills

English (first language), Finnish (fluent), Spanish (excellent), Japanese (beginning)